

Dhanavath Balaji

+91-9059827348 | balajidhan1301@gmail.com | LinkedIn | Github

EDUCATION

Indian Institute of Technology Guwahati, India

Bachelor of Technology in Mechanical Engineering

Aug 2019 - May 2023

EXPERIENCE

•HCL Tech, Hyderabad

Oct 2023 - Jan 2026

Technical Lead

- Architected scalable backend services for **Volvo racks** using **C# and ASP.NET Core Web API**, integrating **SQL Server** to implement standardized rack mapping logic, improving operational efficiency by **20%**.
- Developed **RESTful APIs** for LPN scanning validation and real-time serial number processing (**5,000+** units/-month), implementing structured exception handling and optimized database queries.
- Engineered modular backend components following **SOLID principles and layered architecture**, reducing manual rack verification efforts by **30%** and improving maintainability.
- Built a real-time defect management system using **ASP.NET Core and MSSQL**, enabling automated defect logging, intelligent rework routing, and increasing defect closure rate by **40%**.
- Designed and implemented **responsive UI** using **HTML, CSS, and JavaScript** integrated with backend APIs, reducing manual errors by **35%** and supporting **50+** daily operators.
- **Tech Stack** : C#, ASP.NET Core, REST APIs, SQL Server, HTML, CSS, JavaScript
- Engineered the **Cummins-Meritor** OEE system, developing the Admin interface for operator management and workflow configuration, and the OEE Dashboard for real-time OEE tracking and machine performance monitoring across hierarchical MFG levels.
- Enhanced traceability and production efficiency by implementing server-driven data processing and user-friendly dashboards using **React.js** interfaced with MES services via **REST APIs**, leveraging **SQL Server** for backend data management, and **Python** scripts within Ignition for real-time automation to ensure accurate machine availability, performance, and quality metrics for OEE analysis.
- Achieved **15%** boost in operational efficiency and a **20%** reduction in downtime.
- **Tech Stack** : Python, MSSQL, React.js

PROJECTS

•Replit

- Designed and developed an **Online Code Editor** platform to simplify project initiation in popular frameworks. Introduced a secure remote code execution environment through **Docker**, enabling users with shell access with enhanced coding efficiency while maintaining stringent security standards.
- Implemented a feature to streamline frontend deployment using **AWS S3 and Redis queues**, enhancing automation.
- **Tech Stack** : Node.js, Socket.io

•E-Wallet

- Constructed a secure and scalable e-wallet system following **microservice** architecture with banking webhook integration to enable seamless real-time transactions.
- Features include user wallet management– Authentication, P2P transactions & history, Balance tracking, and fund transfers. Automated the **CI/CD** pipeline using GitHub Actions for continuous integration and deployment.
- **Tech Stack** : Node.js, TypeScript

TECHNICAL SKILLS

•**Programming Languages:** C#, JavaScript, Python, C++

•**Backend Technologies:** .NET, ASP.NET Core (Web API), RESTful Services

•**Database:** SQL Server (MSSQL), MySQL, MongoDB

•**Frontend:** HTML, CSS, JavaScript, React.js, Angular

•**Cloud Platforms & Tools:** Git, GitHub, Azure (Basics), AWS, Docker, VS Code, Postman, Jupyter

•**System Design:** High-Level Design (HLD), Object-Oriented Design, SOLID Principles, Low-Level Design (LLD), Clean Architecture

CERTIFICATIONS

• **Certifications:** **Gen AI** Consumer Training, **Gen AI** Prompt engineering, **Data Structures and Algorithms** in C++, C# Programming, **Python** Developer in **HackerRank**.

ACHIEVEMENTS AND EXTRA CURRICULAR

- **Technical Lead, HCL Tech** :Drove a cross-functional team of 15 members, streamlining task coordination and project tracking, resulting in a 21% improvement in work efficiency and on-time project delivery.
- **Core Team Member, Alcheringa IITG**:Managed participant engagement and Spearheaded branding efforts for City Elimination and Alcher Got Talent events, including Bangalore's digital and city branding for Alchering.